

```
In [1]: # Multiple Linear Regression
        # Profit Prediction using MLR

        # CORE WORDS- Categorical data types 1.Nominal data 2. Ordinal data
        # Nominal data -One Hot Encoding -Number of Columns will increase
        # Ordinal data- Mapping-Label Encoder-no column expansion

        # Prefer this algorithm when more than one input, datas are in numbers

import pandas as pd
```

```
In [2]: # DATA COLLECTION

dataset=pd.read_csv("50_Startups.csv")
```

```
In [3]: # Dataset has categorial column(nominal datas) (text column ex:name of a pe
        # rson,place,thing)
        # categorical column is coverted to numerical column by one hot encoding me
        # thod,by this column will be expands for each category each column.

        # Converting Nominal data into numerical data by using get_dummies

        # syntax: dataset=pd.get_dummies(dataset)
        #          dataset

dataset
```

Out[3]:

	R&D Spend	Administration	Marketing Spend	State	Profit
0	165349.20	136897.80	471784.10	New York	192261.83
1	162597.70	151377.59	443898.53	California	191792.06
2	153441.51	101145.55	407934.54	Florida	191050.39
3	144372.41	118671.85	383199.62	New York	182901.99
4	142107.34	91391.77	366168.42	Florida	166187.94
5	131876.90	99814.71	362861.36	New York	156991.12
6	134615.46	147198.87	127716.82	California	156122.51
7	130298.13	145530.06	323876.68	Florida	155752.60
8	120542.52	148718.95	311613.29	New York	152211.77
9	123334.88	108679.17	304981.62	California	149759.96
10	101913.08	110594.11	229160.95	Florida	146121.95
11	100671.96	91790.61	249744.55	California	144259.40
12	93863.75	127320.38	249839.44	Florida	141585.52
13	91992.39	135495.07	252664.93	California	134307.35
14	119943.24	156547.42	256512.92	Florida	132602.65
15	114523.61	122616.84	261776.23	New York	129917.04
16	78013.11	121597.55	264346.06	California	126992.93
17	94657.16	145077.58	282574.31	New York	125370.37
18	91749.16	114175.79	294919.57	Florida	124266.90
19	86419.70	153514.11	0.00	New York	122776.86
20	76253.86	113867.30	298664.47	California	118474.03
21	78389.47	153773.43	299737.29	New York	111313.02
22	73994.56	122782.75	303319.26	Florida	110352.25
23	67532.53	105751.03	304768.73	Florida	108733.99
24	77044.01	99281.34	140574.81	New York	108552.04
25	64664.71	139553.16	137962.62	California	107404.34
26	75328.87	144135.98	134050.07	Florida	105733.54
27	72107.60	127864.55	353183.81	New York	105008.31
28	66051.52	182645.56	118148.20	Florida	103282.38
29	65605.48	153032.06	107138.38	New York	101004.64
30	61994.48	115641.28	91131.24	Florida	99937.59
31	61136.38	152701.92	88218.23	New York	97483.56
32	63408.86	129219.61	46085.25	California	97427.84
33	55493.95	103057.49	214634.81	Florida	96778.92
34	46426.07	157693.92	210797.67	California	96712.80
35	46014.02	85047.44	205517.64	New York	96479.51
36	28663.76	127056.21	201126.82	Florida	90708.19
37	44069.95	51283.14	197029.42	California	89949.14

	R&D Spend	Administration	Marketing Spend	State	Profit
38	20229.59	65947.93	185265.10	New York	81229.06
39	38558.51	82982.09	174999.30	California	81005.76
40	28754.33	118546.05	172795.67	California	78239.91
41	27892.92	84710.77	164470.71	Florida	77798.83
42	23640.93	96189.63	148001.11	California	71498.49
43	15505.73	127382.30	35534.17	New York	69758.98
44	22177.74	154806.14	28334.72	California	65200.33
45	1000.23	124153.04	1903.93	New York	64926.08
46	1315.46	115816.21	297114.46	Florida	49490.75
47	0.00	135426.92	0.00	California	42559.73
48	542.05	51743.15	0.00	New York	35673.41
49	0.00	116983.80	45173.06	California	14681.40

In [4]: *# PREPROCESSING*

To avoid repeted data and to consume memory drop_first =True means removing the first column of the converted numerical column
here state_california is removed since it is first column

```
dataset=pd.get_dummies(dataset,drop_first=True)  
dataset
```

Out[4]:

	R&D Spend	Administration	Marketing Spend	Profit	State_Florida	State_New York
0	165349.20	136897.80	471784.10	192261.83	0	1
1	162597.70	151377.59	443898.53	191792.06	0	0
2	153441.51	101145.55	407934.54	191050.39	1	0
3	144372.41	118671.85	383199.62	182901.99	0	1
4	142107.34	91391.77	366168.42	166187.94	1	0
5	131876.90	99814.71	362861.36	156991.12	0	1
6	134615.46	147198.87	127716.82	156122.51	0	0
7	130298.13	145530.06	323876.68	155752.60	1	0
8	120542.52	148718.95	311613.29	152211.77	0	1
9	123334.88	108679.17	304981.62	149759.96	0	0
10	101913.08	110594.11	229160.95	146121.95	1	0
11	100671.96	91790.61	249744.55	144259.40	0	0
12	93863.75	127320.38	249839.44	141585.52	1	0
13	91992.39	135495.07	252664.93	134307.35	0	0
14	119943.24	156547.42	256512.92	132602.65	1	0
15	114523.61	122616.84	261776.23	129917.04	0	1
16	78013.11	121597.55	264346.06	126992.93	0	0
17	94657.16	145077.58	282574.31	125370.37	0	1
18	91749.16	114175.79	294919.57	124266.90	1	0
19	86419.70	153514.11	0.00	122776.86	0	1
20	76253.86	113867.30	298664.47	118474.03	0	0
21	78389.47	153773.43	299737.29	111313.02	0	1
22	73994.56	122782.75	303319.26	110352.25	1	0
23	67532.53	105751.03	304768.73	108733.99	1	0
24	77044.01	99281.34	140574.81	108552.04	0	1
25	64664.71	139553.16	137962.62	107404.34	0	0
26	75328.87	144135.98	134050.07	105733.54	1	0
27	72107.60	127864.55	353183.81	105008.31	0	1
28	66051.52	182645.56	118148.20	103282.38	1	0
29	65605.48	153032.06	107138.38	101004.64	0	1
30	61994.48	115641.28	91131.24	99937.59	1	0
31	61136.38	152701.92	88218.23	97483.56	0	1
32	63408.86	129219.61	46085.25	97427.84	0	0
33	55493.95	103057.49	214634.81	96778.92	1	0
34	46426.07	157693.92	210797.67	96712.80	0	0
35	46014.02	85047.44	205517.64	96479.51	0	1
36	28663.76	127056.21	201126.82	90708.19	1	0
37	44069.95	51283.14	197029.42	89949.14	0	0

	R&D Spend	Administration	Marketing Spend	Profit	State_Florida	State_New York
38	20229.59	65947.93	185265.10	81229.06	0	1
39	38558.51	82982.09	174999.30	81005.76	0	0
40	28754.33	118546.05	172795.67	78239.91	0	0
41	27892.92	84710.77	164470.71	77798.83	1	0
42	23640.93	96189.63	148001.11	71498.49	0	0
43	15505.73	127382.30	35534.17	69758.98	0	1
44	22177.74	154806.14	28334.72	65200.33	0	0
45	1000.23	124153.04	1903.93	64926.08	0	1
46	1315.46	115816.21	297114.46	49490.75	1	0
47	0.00	135426.92	0.00	42559.73	0	0
48	542.05	51743.15	0.00	35673.41	0	1
49	0.00	116983.80	45173.06	14681.40	0	0

In [5]: `dataset.columns`

Out[5]: `Index(['R&D Spend', 'Administration', 'Marketing Spend', 'Profit',
'State_Florida', 'State_New York'],
 dtype='object')`

In [6]: `#INPUT AND OUTPUT SPLIT UP
independent=INPUT

independent=dataset[['R&D Spend', 'Administration', 'Marketing Spend',
 'State_Florida', 'State_New York']]`

In [7]: independent

Out[7]:

	R&D Spend	Administration	Marketing Spend	State_Florida	State_New York
0	165349.20	136897.80	471784.10	0	1
1	162597.70	151377.59	443898.53	0	0
2	153441.51	101145.55	407934.54	1	0
3	144372.41	118671.85	383199.62	0	1
4	142107.34	91391.77	366168.42	1	0
5	131876.90	99814.71	362861.36	0	1
6	134615.46	147198.87	127716.82	0	0
7	130298.13	145530.06	323876.68	1	0
8	120542.52	148718.95	311613.29	0	1
9	123334.88	108679.17	304981.62	0	0
10	101913.08	110594.11	229160.95	1	0
11	100671.96	91790.61	249744.55	0	0
12	93863.75	127320.38	249839.44	1	0
13	91992.39	135495.07	252664.93	0	0
14	119943.24	156547.42	256512.92	1	0
15	114523.61	122616.84	261776.23	0	1
16	78013.11	121597.55	264346.06	0	0
17	94657.16	145077.58	282574.31	0	1
18	91749.16	114175.79	294919.57	1	0
19	86419.70	153514.11	0.00	0	1
20	76253.86	113867.30	298664.47	0	0
21	78389.47	153773.43	299737.29	0	1
22	73994.56	122782.75	303319.26	1	0
23	67532.53	105751.03	304768.73	1	0
24	77044.01	99281.34	140574.81	0	1
25	64664.71	139553.16	137962.62	0	0
26	75328.87	144135.98	134050.07	1	0
27	72107.60	127864.55	353183.81	0	1
28	66051.52	182645.56	118148.20	1	0
29	65605.48	153032.06	107138.38	0	1
30	61994.48	115641.28	91131.24	1	0
31	61136.38	152701.92	88218.23	0	1
32	63408.86	129219.61	46085.25	0	0
33	55493.95	103057.49	214634.81	1	0
34	46426.07	157693.92	210797.67	0	0
35	46014.02	85047.44	205517.64	0	1
36	28663.76	127056.21	201126.82	1	0
37	44069.95	51283.14	197029.42	0	0

	R&D Spend	Administration	Marketing Spend	State_Florida	State_New York
38	20229.59	65947.93	185265.10	0	1
39	38558.51	82982.09	174999.30	0	0
40	28754.33	118546.05	172795.67	0	0
41	27892.92	84710.77	164470.71	1	0
42	23640.93	96189.63	148001.11	0	0
43	15505.73	127382.30	35534.17	0	1
44	22177.74	154806.14	28334.72	0	0
45	1000.23	124153.04	1903.93	0	1
46	1315.46	115816.21	297114.46	1	0
47	0.00	135426.92	0.00	0	0
48	542.05	51743.15	0.00	0	1
49	0.00	116983.80	45173.06	0	0

In [8]: `#dependent=OUTPUT`
`dependent=dataset[["Profit"]]`

In [9]: dependent

Out[9]:

	Profit
0	192261.83
1	191792.06
2	191050.39
3	182901.99
4	166187.94
5	156991.12
6	156122.51
7	155752.60
8	152211.77
9	149759.96
10	146121.95
11	144259.40
12	141585.52
13	134307.35
14	132602.65
15	129917.04
16	126992.93
17	125370.37
18	124266.90
19	122776.86
20	118474.03
21	111313.02
22	110352.25
23	108733.99
24	108552.04
25	107404.34
26	105733.54
27	105008.31
28	103282.38
29	101004.64
30	99937.59
31	97483.56
32	97427.84
33	96778.92
34	96712.80
35	96479.51
36	90708.19
37	89949.14

	Profit
38	81229.06
39	81005.76
40	78239.91
41	77798.83
42	71498.49
43	69758.98
44	65200.33
45	64926.08
46	49490.75
47	42559.73
48	35673.41
49	14681.40

```
In [10]: # TRAINING AND TEST SET SPLIT UP

from sklearn.model_selection import train_test_split
X_train, X_test, Y_train, Y_test= train_test_split(independent,dependent,te
st_size=0.30,random_state=0)
```

```
In [11]: # LEARNING PHASE
# MODEL CREATION

from sklearn.linear_model import LinearRegression
regressor=LinearRegression()
regressor.fit(X_train,Y_train)
```

```
Out[11]: LinearRegression(copy_X=True, fit_intercept=True, n_jobs=None, normalize=False)
```

```
In [12]: weight=regressor.coef_
weight
```

```
Out[12]: array([[7.90840255e-01, 3.01968165e-02, 3.10148566e-02, 4.63028992e+02,
3.04799573e+02]])
```

```
In [13]: bais=regressor.intercept_
bais
```

```
Out[13]: array([42403.87087053])
```

```
In [14]: #EVALUATION

Y_pred=regressor.predict(X_test)
```

```
In [15]: from sklearn.metrics import r2_score
r_score=r2_score(Y_test,Y_pred)
```

```
In [16]: r_score
```

```
Out[16]: 0.9358680970046241
```

```
In [ ]: # TO SAVE MODEL
```

```
import pickle  
filename="finalized_model_multiple_linear.sav"
```

```
In [ ]: pickle.dump(regressor,open(filename,'wb'))
```

```
In [ ]: # TO LOAD THE MODEL FOR CHECKING
```

```
loaded_model=pickle.load(open("finalized_model_multiple_linear.sav",'rb'))  
result=loaded_model.predict([[1234,345,346,1,0]])
```

```
In [ ]: result
```

```
In [ ]:
```